

Software Requirement Specification Document

Phishing Detection System

Group 05

1. INTRODUCTION

1.1 Purpose

So, the purpose of this Software Requirements Specification SRS, is to lay out the functional and non-functional requirements for the Phishing Detection System. Basically, this document goes through what the system must do, what it must be able to support, and what limitations it has. It also gives a detailed description of the system's features, services, constraints, and expected behavior. In the end it is meant to be a communication link between stakeholders, developers, testers, and project supervisors for the whole development life cycle.

The Phishing Detection System is meant to help users spot potentially malicious websites and phishing URLs before they click or interact with anything. The system inspects URLs using rule-based phishing detection techniques, then it computes a phishing risk score from predefined security cues. The outcome is shown to users in a web application and also via a browser extension. This SRS should create a solid understanding of what the system should accomplish, and it also becomes a starting point for the system design, implementation, testing, deployment, and maintenance. The document stays mostly on the system requirements and the expected functions, not on the technical implementation details, because that's usually handled later.

1.2 Scope

The Phishing Detection System is a web based cyber security system built with Python & browser extension. The main goal of the system is to help the users detect the phishing websites based on the characteristics of URL and to provide a risk score calculation.

The system can be used to manually add URLs via a web interface, or manually scan the currently visited website using the browser extension. It checks different phishing markers like the presence of suspicious characters, too long of a URL, lack of a secure connection, multiple subdomains, and the use of IP addresses instead of domain names. According to the information it detects, it will calculate a risk rating and determine if the website is Safe, Suspicious or Phishing.

The browser extension allows users to manually scan the currently visited website while browsing the internet. When the user activates the extension and clicks the scan button, the extension captures the current webpage URL and submits it to the back-end server for analysis. The user gets a result as a risk score and warning message. The system also keeps a data base of scanned URL and their analysis data. Previous scan records can be viewed by users and system activities can be monitored and scan statistics can be reviewed via an admin dashboard.

The scope of the system includes:

- URL scanning and analysis.
- Rule-based phishing detection.
- Phishing risk score calculation.
- Browser extension phishing warning after manual scanning.
- Storage of scan history and detection results.
- User-friendly web interface.
- Administrator monitoring and reporting features.

The scope of the project does not include:

- Machine learning-based phishing prediction.
- Email phishing detection.
- Mobile application support.
- Integration with external threat intelligence APIs.
- Automatic blocking of websites.

The system is intended to provide users with awareness and assistance in identifying potentially dangerous websites and reducing the risk of phishing attacks.

1.3 Definitions and Acronyms

The following definitions and acronyms are used throughout this document.

Term/Acronym	Definition
SRS	Software Requirements Specification
URL	Uniform Resource Locator, commonly known as a website address
UI	User Interface
Admin	Administrator responsible for monitoring system activities
Browser Extension	A software component that extends browser functionality and enables real-time URL scanning
Phishing	A cybercrime technique that uses deceptive websites or messages to steal sensitive user information
Risk Score	A numerical value representing the likelihood that a website is phishing-related
Rule-Based Detection	A detection method that identifies phishing websites using predefined rules and conditions
Backend	The server-side component responsible for processing requests and performing URL analysis
Frontend	The user-facing component that allows interaction with the system
Database	A structured collection of data used to store scanned URLs and detection results
MongoDB	The database management system used to store phishing detection records
HTTPS	Hypertext Transfer Protocol Secure, a secure communication protocol used on websites
Domain Name	The human-readable address of a website such as example.com
Scan History	A collection of previously analyzed URLs and their detection results
Detection Engine	The module responsible for analyzing URLs and calculating phishing risk scores
Real-Time Warning	An immediate notification displayed when a suspicious or phishing website is detected

2. OVERALL DESCRIPTION

2.1 Product Perspective

Phishing Detection System is a self-contained web-based cyber security application to identify potentially malicious websites based on URL analysis. The system employs phishing detection methods based on rules, and offers phishing risk assessment services via web app and browser extension.

The system can be divided into five major components:

1. User Interface
2. Browser Extension
3. Python Backend Server
4. Phishing Detection Engine
5. MongoDB Database

Users can type in URLs via the web application, or manually scan the currently visited website using the browser extension. When the user activates the browser extension and click the scan button, the extension capture the URL of the currently visited website and sends it to the back-end server for analysis.

The URL is passed to the backend server, which runs the phishing detection engine. The detection engine analyzes a number of phishing markers, including number of subdomains, no secure protocols, suspicious symbols, excessive URL length, and IP addresses, among others. In the result of the analysis, a phishing risk score is created.

The risk score, phishing status and explanation are returned to the user via the web interface or browser extension. The scan responses are recorded in the database for future reference and administrative control.

It's a standalone solution that doesn't use machine learning models, external threat intelligence sources, or third-party phishing detection APIs.

2.2 Product Functions

The Phishing Detection System offers a number of features to help users detect phishing sites and raise their awareness of cyber security.

URL Submission: Users can submit web URLs via the web interface for phishing analysis in the system.

URL Feature Extraction: The system retrieves the relevant URL attributes that can be used to detect phishing activities such as URL length, unusual characters, subdomain structure, protocol, domain characteristics etc.

Rule-Based URL Analysis: The system compares the URL features extracted to the predefined phishing detection rules.

Phishing Risk Score Generation: The system returns a percentage for the phishing risk, depending on the amount and severity of the phishing indicators found.

Detection Result Display: The system shows if the scanned URL is Safe, Suspicious or Phishing.

Detection Explanation: The system gives explanations when explaining the reasons for phishing scores generated.

Browser Extension Scanning: The extension allows users to manually scan the currently visited website. When activated, the extension captures the current URL and sends it to the backend server for analysis.

Warning notifications after scan: A warning message is issued when suspicious or phishing websites are identified.

Scan History Management: The system can be set up so that previously scanned URLs with analysis results can be viewed again.

Administrative Monitoring: The system enables administrators to track scanning activities, view scan records, and gain insights into usage trends.

Database Management: The system stores the URLs, phishing scores, phishing detection results, timestamps, and scan history record in the database.

2.3 User Classes and Characteristics

The Phishing Detection System has two main groups of users.

General User

General users: People who use the system to analyze URLs and determine whether the site is a phishing site or not.

Characteristics of General Users:

- Basic computer literacy.
- Basic web browsing experience.
- No need for cyber security expertise.
- Interested in knowing which websites to avoid sharing personal information with.

General Users' responsibilities:

- Input URLs to be analyzed. See phishing risk scores.
- Review detection explanations.
- View the history of scans.
- Use the Browser Extension while browsing.

Administrator

The administrators are tasked to monitor system operations and ensuring the effectiveness of the system.

Characteristics of Administrators:

- Knowledge of system operation.
- Basic technical knowledge.
- Accountable to monitor system activities.

Responsibilities of Administrators:

- Monitor scan records.
- View system statistics.
- Check phishing detection results.
- Maintain database records.
- Create reports as needed.

2.4 Operating Environment

The Phishing Detection System will operate within the following software and hardware environment.

Hardware Environment

- Computer or Laptop
- Intel Core i5 Processor or higher
- Minimum 8 GB RAM
- Minimum 20 GB available storage
- Stable internet connection

Software Environment

- Operating Systems:
 - Microsoft Windows
 - Linux
 - macOS
- Programming Language:
 - Python
- Frontend Technologies:
 - HTML
 - CSS
 - JavaScript
- Database:
 - MongoDB
- Development Environment:
 - Visual Studio Code
- Version Control:
 - GitHub
- Browser:
 - Google Chrome

Browser Extension Environment

The browser extension will operate within Google Chrome and communicate with the backend server through HTTP requests.

2.5 Assumptions and Constraints

Assumptions

During the system development and operating, the following assumptions are considered:

- Users are able to access a fixed broadband network.
- Users are able to use a supported Web browser.
- Users submit legitimate URLs of web sites for analysis.
- The back-end server is still functioning and accessible.
- MongoDB services for database continue to be available.
- Users are familiar with the basics of navigating the Web.

Constraints

The system has the following restrictions:

- The system only relies on phishing detection techniques based on rules.
- Machine learning algorithms are not used.
- No external phishing detection APIs are used.
- Only URLs are analyzed; not in detail the web content.
- The browser extension is created with the Google Chrome browser in mind.
- The accuracy of detection is reliant on the rules for phishing which are defined.
- The system does not automatically block websites, it is used to give recommendations and warnings.
- The system must have internet connection for communication between the Browser extension and the Backend server.

3. USER REQUIREMENTS

3.1 URL Scanning Requirement

The system should be able to input a website URL via the web application to be analyzed for phishing. It should be easy for users to submit URLs, without needing technical expertise, through an easy-to-use interface. Once a URL is submitted, the system shall process the request and analyze for phishing.

The system should be capable of analyzing the most popular URL of websites and should give back the result within a reasonable time. Users are allowed to scan several URLs at a time and get result for each URL.

3.2 Phishing Risk Assessment Requirement

The system shall deliver users an evaluation of the phishing risk of the scanned URL. The assessment shall be displayed as a percentage of risk for the website being malicious or suspicious.

The system shall provide a measure to assess the risk level of a given URL based on one or more phishing detection rules. The risk assessment shall help users to make informed decisions prior to using websites.

3.3 Browser Extension Requirement

The system shall offer a browser add-on to help the users determine if a web site may pose a risk when surfing the web.

The browser extension shall capture the URL for the currently visited website when the user activates the extension and initiates a scan. A phishing risk information will be delivered directly to users via the extension without having to manually type the URLs into the web application.

The extension shall provide phishing analysis on demand whenever the user requests a scan and will offer easy access to phishing detection when doing normal surfing.

3.4 Detection Result Requirement

The system shall display the phishing detection results in an understandable and clear manner. Users will be notified of the website scanned as Safe, Suspicious or Phishing.

The results of the detection shall contain a phishing risk score and the explanation of the various factors that went into the assessment. The information shall be presented in a way that is comprehensible to users with limited knowledge of cyber security.

The system shall send warnings if a site is determined to be potentially harmful.

3.5 Scan History Requirement

The system shall keep a list of previously scanned URLs and results of the scans.

Users should be able to access and view their scan history, to see all the websites that have been scanned previously. The scan history should contain details like scanned URL, phishing risk score, phishing detection, analysis date etc.

A history of the scans shall be securely stored within the system and shall be accessible when needed.

3.6 Administrator Monitoring Requirement

The system shall be capable of administration for monitoring phishing detection activities and system usage.

Admins will be able to check scan logs, analyze phishing detection and statistics. The system shall deliver data to the administrators on the number of scanned URLs, number of detected phishing sites, and level of activity in the system.

System components of administrative monitoring shall be used for system maintenance, performance evaluation and management of stored detection records.

4. SYSTEM REQUIREMENTS

4.1 Functional Requirements

4.1.1 Functional Requirement 1 – URL Input

Description:

The system should enable the user to enter the URL to be analyzed for phishing using the web application.

Input:

The address for the web page that the user types in.

Process:

The system checks the URL that was entered in the box and passes it on to the phishing detection engine for analysis.

Output:

The following URL was validated:

4.1.2 Functional Requirement 2 – URL Feature Extraction

Description:

The system shall be able to identify phishing properties based on the provided URL.

Input:

Validated URL.

Process:

The system processes attributes of URL such as:

- URL length
- Special characters present
- Number of subdomains
- Use of IP address instead of domain name

Having or not having HTTPS. Whether HTTPS exists or not.

- Suspicious keywords

Output:

Extracted URL features.

4.1.3 Functional Requirement 3 – URL Analysis

Description:

The system shall cross reference the extracted URL features with the predetermined rules for detecting phishing.

Input:

Extracted URL features.

Process:

The system performs a comparison between the characteristics of the URL against the phishing rules that are based on indicators and detects the suspicious behavior.

Output:

Indicators of phishing found in analysis result.

4.1.4 Functional Requirement 4 – Risk Score Generation

Description:

The system shall output a risk score of the analyzed URL for phishing.

Input:

Analysis result.

Process:

The system scores the phishing indicators that are identified, and gives them weights to determine the total risk percentage.

Output:

A percentage that represents the level of phishing risk that an Internet browser has encountered.

4.1.5 Functional Requirement 5 – Detection Result Display

Description:

The system shall show the result of phishing detection to users.

Input:

Risk score.

Process:

The system segments the website into pre-defined risk ranges.

Output:

Safe and Suspicious/PHISHING status with the risk score.

Example:

- 0% – 30% = Safe
- 31% – 60% = Suspicious
- 61% – 100% = Phishing

4.1.6 Functional Requirement 6 – Browser Extension Scanning

Description:

The system shall enable automatic scanning of URLs in a browser by means of browser extension.

Input:

Currently visited website URL.

Process:

When the user clicks the Scan button in the browser extension, the extension captures the current webpage URL and sends it to the backend server for analysis.

Output:

New Risk Score and Phishing status are shown in the browser extension.

4.1.7 Functional Requirement 7 – Real-Time Warning Notification

Description:

If a suspicious or phishing website is detected, the system shall notify the users.

Input:

Detection result.

Process:

System includes risk assessment and provides warning when risk exceeds pre-defined limits.

Output:

A message to alert the user to some problem.

4.1.8 Functional Requirement 8 – Scan History Management

Description:

The system shall be able to store a record of URLs scanned.

Input:

The following are displayed: URL, risk score, and detection result.

Process:

System stores scan data and retrieves previous scan data.

Output:

User-accessible scan history.

4.1.9 Functional Requirement 9 – Database Storage

Description:

The system should maintain a database of phishing detection information.

Input:

Scan details.

Process:

URL information, risk score, detection status, explanations and timestamps are stored.

Output:

Long-term storage of phishing detection information.

4.1.10 Functional Requirement 10 – Administrator Dashboard

Description:

System shall provide an administration dashboard.

Input:

Stored scan records.

Process:

The system provides a summary and statistics on the scanning activities.

Output:

Reports and information on monitoring.

4.1.11 Functional Requirement 11 – Detection Explanation Generation

Description:

The system shall offer explanations of phishing scores generated.

Input:

Detected phishing indicators.

Process:

The system detects the rules fired and provides human-readable explanations.

Output:

An explanation of the reasons for the assigned Risk Score.

Example:

- URL contains suspicious special characters.
- URL is not secured with HTTPS.

The URL of the page is too long.

4.2 Non-Functional Requirements

4.2.1 Performance Requirements

Within 2 seconds, in normal operation, the system shall display the result of phishing detection. The extension to the browser should report the detection results as soon as possible. Multiple scan requests should be accepted and processed in an efficient manner without significant performance impact.

4.2.2 Security Requirements

The system shall ensure that data stored is not accessible to unauthorized users. No other users shall have access to administrative functions. The system shall be able to securely handle communication between the browser extension and backend server. The system shall not allow unauthorized modification of the records of a scan that are stored.

4.2.3 Reliability Requirements

The system shall be able to reliably and consistently report phishing detection results according to the rules implemented. The system shall function normally without unexpected failures during normal operation. The system shall be graceful in responding to temporary errors and interruptions.

4.2.4 Usability Requirements

The user interface shall be simple to use and easy to understand. Users will be able to perform URL scans without special technical expertise. The results of the detection shall be clearly reported in terms that can be understood. Warning messages should be readily understandable.

4.2.5 Availability Requirements

The system shall be run when the internet connection and server resources are available. The web application shall be accessible to the user continuously via the standard web browser. The browser extension should not be closed when the browser is closed.

4.2.6 Scalability Requirements

As more and more URL scans are used, the system shall be able to support them more.

The database must be able to hold a large number of scan records.

The architecture shall enable expansion of phishing detection rules in the future.

4.2.7 Maintainability Requirements

Future changes and upgrades to the system are to be supported.

New phishing detection rules will be added to the system without a significant modification of the system architecture.

Source code will be structured and documented for ease of maintenance.

The system should be structured in a modular fashion, making debugging and improving the system easier.

5. EXTERNAL INTERFACE REQUIREMENTS

5.1 User Interface Requirements

The Phishing Detection System shall produce a user-friendly web interface that will allow users to interact with the system efficiently. The interface will be easy to read, easy to understand and easy to respond to, for users of varying technical expertise

5.1.1 Home Page

The Home Page will be the first page of the system. The Home Page shall include:

- A short introduction to the phishing detection system.
- Links to other pages of the system.
- A URL box for instant access to phishing analysis.
- Information on how the system works and what it is used for in terms of purpose.
- A history of scans and administrative options (if applicable).

The page should be attractive and user-friendly.

5.1.2 URL Scan Page

The URL Scan Page will enable users to input URLs for phishing analysis.

The page must have the following:

- URL input field.
- Scan button.
- Input validation messages.
- Properly entering URLs.

Users should be able to type in a URL and click to start a scan.

Processing indicators shall be displayed on the page during the analysis process.

5.1.3 Result Page

The Result Page shall show the analysis results of phishing. The page will display:

- Submitted URL.
- Phishing Risk Score in percentage.
- Websites' classification status (Safe, Suspicious, or Phishing).
- Detection explanation.
- Date and time of analysis.

The results should be presented in an understandable and visible manner, so the user can make decisions regarding security.

5.1.4 Scan History Page

The user shall be able to view URLs previously scanned on the Scan History Page.

The web page should contain the following text:

- Scanned URL.
- Risk score.
- Detection result.
- Date and time of scan.

Users should have access to a browsable history of scan records and the results of previous scans.

5.1.5 Administrator Dashboard

The role of the Administrator Dashboard is to offer administrative monitoring capabilities.

The dashboard should show:

- Total scanned URLs:
- Total number of scanned URLs.
- Number of phishing Web sites detected.
- The number of websites that you can use safely.
- Recent scanning activities.
- Section scan averages and summary statistics.

Administrators can view and track system usage and detection history on this interface.

5.2 Browser Interface Requirements

At the same time, the system shall contain a browser add-on that will enable the system to allow users to manually scan websites while browsing the web.

5.2.1 Browser Extension Popup

The browser extension shall have a popup user interface that will show the phishing detection information.

This popup will display:

- Current website URL.
- Risk score percentage.
- Detection status.
- Detection explanation.
- Scan Current website button.

The users should be able to view the analysis results of phishing attack directly from the extension pop up.

5.2.2 Browser Notification Interface

The browser extension will alert users if suspicious websites are found.

The notification interface will:

- Display warning messages.
- Mark Phishing risk level.
- Warn users prior to engaging with questionable websites.

The notifications shall be visible and easily comprehensible

5.2.3 URL Capture Interface

The browser extension shall capture the URL of the currently viewed webpage when the user initiates a scan.

The URL capture interface shall:

- Capture the URL of the currently active webpage
- Send the URL to the backend server for analysis
- Receive and display the phishing analysis results.
- Get the Web address of the current page.
- Send the URL to the back-end server for analysis.

The process of capturing URLs shall begin only when the user activates the browser extension and clicks the scan button.

5.3 Database Interface Requirements

The system shall use the MongoDB database for storing and fetching phishing detection information.

5.3.1 Database Connection

The system will connect to the MongoDB Database securely.

The database connection will:

- Support data insertion.
- Support data retrieval.
- Ensure to keep the connection reliable in normal operation.

The entire database communication shall be handled via the backend server.

5.3.2 URL Data Storage

The system shall be able to save scanned URL data into the database.

The information to be stored shall contain:

- URL address.
- Risk score.
- Detection status.
- Detection explanation.
- Scan timestamp.

The system shall ensure the integrity of data while storing data.

5.3.3 Scan History Retrieval

The system shall be able to retrieve stored scan records from the database on request.

The following may be allowed during the retrieval process:

- Said that they had had a look at the scan history.
- Accessing historical detection results.
- Generating administrative reports.

The system shall be able to provide accurate and complete records.

5.3.4 Risk Score Storage

The system shall record the phishing risk scores computed based on the analysis of URLs.

The information shall be entered into the storage and shall contain:

- Risk percentage score.
- Detection category.
- Associated URL.
- Check the date & time.

The data collected shall be available for future retrieval and reporting.

6. SYSTEM ARCHITECTURE

6.1 Architecture Overview

The Phishing Detection System is designed with client-server architecture, which includes a web application, a browser extension, a Python-based back-end server, a phishing detection engine, and a mongo database. The architecture is designed to be able to detect phishing on the fly, both by manually scanning URLs and by manually scanning through the browser extension.

The interaction between the user and the system is the web application or the browser extension. The user enters a URL into the browser, which is passed to the Python backend server, fed through the phishing detection engine and added to the MongoDB database. Risk score and detection result generated for the phishing are returned to the user.

6.2 User Interface Component

The User Interface component is the first place where user and phishing detection system interact.

The User Interface's functions include:

- URL submission.
- Showing phishing risk ratings.
- Showing the results of phishing detection.
- Presenting detection explanations.
- Viewing scan history.
- Accessing administrative functions.

The user interface is created with HTML, CSS and JavaScript.

6.3 Browser Extension Component

The Browser Extension component allows to detect phishing in real time while surfing the Internet.

The Browser Extension has the following functions:

- Capturing the URL of the currently visited website when requested by the user.
- Uploading links to the back-end server.
- To receive phishing analysis results.
- Showing phishing risk scores.
- Displaying warning notifications.

6.4 Python Backend Server

The Python Backend Server is the CPU that is at the heart of the system.

The following are considered to be functions of the backend server:

- Receiving scan requests.
- Processing submitted URLs.
- Coordinating the communication between elements of the system.
- Performing phishing detection procedures.
- Retrieving and storing database records.
- Providing feedback to users.

The server in the back-end coordinates with all the major operations of the system.

6.5 Phishing Detection Engine

The Phishing Detection Engine conducts phishing analysis with rule-based detection methods which are pre-defined.

The detection engine analyzes:

- URL length.
- Unusual words or phrases.
- Location of Internet Protocol addresses.
- Number of subdomains.
- HTTPS availability.
- Other pre-determined phishing markers.

The engine analyzes the results of the analysis and calculates the phishing risk score and the website's classification.

6.6 Database Server

All phishing detection records are stored in the MongoDB Database Server.

Stored information includes:

- Website URLs.
- Phishing risk scores.
- Detection results.
- Detection explanations.
- Scan timestamps.

Scan history and administrative reporting is supported by the database.

6.7 System Workflow

The workflow of the system is as follows:

1. User enters a URL in the web supplication or visits a website.
2. If using the browser extension, the user opens the extension and click the Scan button.
3. The browser extension captures the current webpage URL.
4. The URL is sent to Python backend server.
5. URL features are extracted on the back-end server.
6. The phishing detection engine compares the extracted features to the phishing rules.
7. The phishing risk score is determined.
8. The web page is marked as Safe, Suspicious, or Phishing.
9. The outcome is saved in the MongoDB database.
10. User sees the phishing risk score, detection result and explanation.
11. If a phishing site is identified, it will display an advisory message.

7. ACCEPTANCE CRITERIA

The acceptance criteria specify the conditions under which the Phishing Detection System would be acceptable as complete and operating as required by the specified requirements. The system shall be exercised for the following parameters to ensure that all the important functions are working properly.

7.1 URL Scanning Acceptance Criteria

The URL scanning function will be deemed to be acceptable if:

- Users can log in a valid URL in the web interface and get the desired result.
- The system is able to interpret URLs correctly.
- Validated invalid URLs with appropriate validation messages.
- The provided URL is forwarded correctly for analysis.
- More than one URL can be scanned separately.

Test Scenario

Input: <https://www.example.com> Expected

Result: The system receives the URL and begins the Phishing Analysis process.

7.2 Risk Score Acceptance Criteria

The phishing risk score functionality shall be acceptable if:

- Each URL scanned is assigned a phishing risk score by the system.
- The score generated is shown as a percentage.
- The risk score is based on the phishing indicators detected.
- Risk scores vary based upon the characteristics of different URLs.
- The points are correctly shown on the web app as well as the browser addon.

Test Scenario

Input: An email that includes a URL with suspicious phishing attributes.

Expected Result: The system calculates the phishing risk percentage according to the indicators that it detects.

7.3 Browser Extension Acceptance Criteria

Extension capabilities of the browser will be deemed to be acceptable if:

- The Extension is able to successfully retrieve the URL of the webpage being visited.
- The captured URL is forwarded to the back-end server after the user click the scan button.
- The results of the analysis are returned to the extension from the backend.
- The phishing risk score will appear in the extension popup.
- The extension successfully analyzes the current webpage after the user manually initiates a scan.

Test Scenario

Input: User accesses a website with Google chrome and clicks the Scan button in the browser extension.

Expected Result: The extension captures the current webpage URL, sends it for analysis, and display the phishing risk score and detection result.

7.4 Warning Notification Acceptance Criteria

The warning notification functionality shall be deemed to be accepted when:

- The system informs of suspicious and phishing websites.
- Warning messages clearly inform about possible security threat.
- Notifications are shown immediately after the analysis.
- Warning messages are easily understood by non-technical users.

Test Scenario

Input: The URL is deemed to be phishing.

Expected Result: A warning notification is shown to the user that indicates the threat that is being detected.

7.5 Scan History Acceptance Criteria

The scan history functionality shall be deemed to be acceptable if:

All scans are saved. Previously scanned URLs can be accessed by users.

In the Stored Records section, the following details are recorded:

- URL, Risk Score, Detection Result, and Scan Date.
- History records are accurately retrieved from the database.
- Previous records are still available after a restart of the system.

Test Scenario

Input: A series of URL scans by a user.

Expected Result: Scan history shows all the scan records correctly.

7.6 Administrator Dashboard Acceptance Criteria

On the administrator "dashboard" functionality shall be deemed acceptable if:

- Administrators can log in to the dashboard.
- The dashboard shows you the record of scans stored.
- Correct system stats are shown on the dashboard.
- Administrators can track phishing detection.
- New URLs that are scanned are included in the statistical information.

Test Scenario

Input: After a number of URL scans, the administrator logs on to the dashboard.

Expected Result: The dashboard provides correct scanning and monitoring information.

Overall System Acceptance

The Phishing Detection System shall be deemed successful if and when:

- All functional requirements are realized.
- All acceptance criteria are met.
- The URL scanning is working properly.
- Phishing risk scores are created correctly.
- Browser extension functionality is working properly.
- Warnings are indicated where necessary.
- History records are scanned and retrieved properly.
- The administrative monitoring features are working correctly.
- No critical errors in any system during the test.
- The system meets the specified non-functional requirements.

8. APPENDICES

8.1 Use Case Diagram

8.2 Use Case Descriptions

8.3 Activity Diagram

8.4 Sequence Diagram

8.5 System Architecture Diagram

8.6 Database Design

8.7 Entity Relationship Diagram (ERD)

8.8 Input and Output Samples

8.9 References